Micro-valence of abstract shapes and colours

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Context and motivation

This study was carried out as part of work package 2 of the EXPERIENCE project (ERC Advanced Grant, PI Axel Cleeremans), which investigates the claim that the phenomenal field is valenced. Lebrecht et al. (2012) argued that everyday objects traditionally considered affectively neutral in affective neuroscience (teapots, chairs, remotes) in fact carry a micro-valence, a subtle affective value triggered automatically upon perception. This micro-valence is proposed to shape how we react to the world around us.

Building on this framework, the present study asked whether low-level visual features, specifically colour, shape and texture, are sufficient to elicit micro-valence in the absence of learned associations with real-world objects.

Stimuli

I designed 96 abstract stimuli as the crossing of 8 saturated colours and 12 shapes. The shapes were selected from Palumbo & Bertamini (2016): 6 curved and 6 angular forms with varying numbers of vertices. Curved shapes are known to be preferred over angular ones (Bar, Neta & Linz, 2006 ; Palumbo & Bertamini, 2016). The 8 colours were drawn from the saturated set in Palmer & Schloss (2010), which captures the strongest variance in human colour preferences according to their ecological valence theory.

Stimuli were named with a letter (A to L) for shape in increasing preference order, and a digit (1 to 8) for colour in increasing preference order. Hence A1 combines the least-preferred shape and colour (greenish, angular, many vertices), whereas L8 combines the most-preferred features (bluish, rounded, few vertices). Each stimulus was a 400 x 400 px image presented on a white background.

Task and procedure

Twenty ULB students participated (17 female, mean age 20.33, SD 3). Protocols were approved by the ULB Ethics Committee and participants gave informed written consent. They were compensated with SONA credits.

The paradigm replicated Lebrecht's (2012) birthday task, designed to probe micro-valence without asking participants to rate valence directly. Participants were shown triplets of stimuli

for 2 seconds and asked either which one they would keep or which one they would return if the triplet had been offered as a birthday gift. Each stimulus appeared 5 times per condition (keep / return), in unique triplets generated in Python, with the constraint that no shape or colour appeared twice in the same triplet. If participants chose randomly, micro-valence scores would cluster around zero ; if they perceived consistent valence, scores would be systematically positive in one condition and negative in the other.

Analysis

I computed a micro-valence score per stimulus per participant by adding 1 for each selection in the keep condition and subtracting 1 for each selection in the return condition. Possible scores ranged from -5 to +5. I then computed group-level means and 95% confidence intervals in JASP. A stimulus was considered to have significant micro-valence when its 95% CI did not cross zero.

Results

Participants were highly consistent. Of 20 participants, 15 to 18 reached extreme micro-valence scores of plus or minus 4 or plus or minus 5 on individual stimuli, comparable to Lebrecht's original figures (17 to 19 out of 20).

Group-level means ranged from +2.00 to -1.73, compared with +2.30 to -2.54 in Lebrecht (2012). The slightly narrower range is consistent with the more abstract nature of my stimuli.

The micro-valence ranking matched previous findings on shape and colour independently. The most-liked stimuli combined bluish colours with rounded shapes ; the least-liked combined greenish colours with angular shapes. These preferences echo the evolutionary hypothesis put forward by Palmer & Schloss (2010) : a preference for blue, associated with clean water and sky, and an aversion to greenish-brown, associated with spoiled food.

Abstract

Throughout the day we make countless implicit decisions: which cup for the morning coffee, which pen to sign a paper, which colour sweater to wear. I argue that visual perception is valenced and directly coupled to affective processing through automatic evaluation mechanisms operating outside awareness. Lebrecht et al. (2015) suggest that objects carry a micro-valence computed by the mind during perception, serving to reduce uncertainty and guide gaze toward some objects rather than others.

While the perception of valence in strongly affective objects is well established, and neutral everyday objects have more recently been shown to carry micro-valence, here I investigated low-level visual features (shape and colour) combined into abstract forms. I conclude that these abstract objects elicit consistent, non-random preferences both at the individual and group level, and therefore carry a micro-valence. With a mild predominance of colour over shape, the results align with previous studies on shape and colour taken separately.

Note (2026) : this pre-doctoral study was my first hands-on experience of the full research pipeline (design, programming in Python, ethics, recruitment, analysis) and my first contact with the EXPERIENCE project's framing of the valenced phenomenal field. The questions it opened (how does the brain assign affective value automatically, and how does this shape perception) still inform my doctoral research on interoception and self-model precision.